



Prospective operational life cycle assessment of energy and waste systems in a university building: an auditable workflow using openLCA and Brightway2

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Abstract

Operational life cycle assessments (LCAs) of campus buildings often use static electricity backgrounds, under-specified scenarios, and limited uncertainty treatment, reducing their usefulness for routine decarbonisation decisions. This study presents a prospective, attributional, location-based workflow for annual building operations. The case study is the Advanced Engineering Building at the University of Queensland, and the functional unit is one building-year. A 2×2 scenario design compares two factors: the electricity background (current Queensland mix versus Queensland-2050) and municipal solid-waste diversion (approximately 64% versus 77%). Operational greenhouse gas impacts are quantified in openLCA as 100-year global warming potential (GWP100; hereafter GWP) using the Australian Life Cycle Inventory database (AusLCI) and the Intergovernmental Panel on Climate Change (IPCC) 2013 method. A lightweight Brightway2 metamodel supports paired contrasts using common random numbers (CRNs) and sensitivity screening. The workflow can be understood as an auditable lifecycle-scenario layer linking a fixed foreground to versioned backgrounds rather than as a cyber-physical twin. Electricity dominates annual operational GWP, and replacing the current electricity background with Queensland-2050 reduces it by about 47%. By contrast, increasing diversion reduces annual operational GWP by only about 5.3 t CO₂-eq yr⁻¹. Sensitivity analysis identifies energy use intensity (EUI) as the dominant driver of variance, whereas waste-related inputs remain minor within the stated scope. The workflow supports repeatable building-level reporting for operators, campus sustainability managers, energy procurement teams, and policy stakeholders. It is reusable across comparable institutional buildings and reporting years.

Keywords Industrial ecology · Prospective life cycle assessment · Operational greenhouse gas emissions · Campus buildings · Scenario analysis · Uncertainty analysis

1 Introduction

Buildings and construction account for a large share of global final energy use and energy-related CO₂ emissions. Recent assessments attribute about one third of both totals to this sector (United Nations Environment Programme & Global Alliance for Buildings and Construction, 2025). Decarbonising the built environment is therefore central to Paris-aligned mitigation pathways. For near-term action, bottom-up studies identify efficiency, electrification, and

cleaner electricity supply as major levers (Camarasa et al., 2022; van Heerden et al., 2025). These priorities motivate operational, decision-oriented life cycle assessments (LCAs) for building management.

However, three methodological limitations still constrain operational LCAs. First, many studies continue to rely on static background assumptions that do not capture time-dependent changes in upstream systems, particularly for electricity (Kordilas et al., 2025; McLaren et al., 2025). For long-lived assets, this can bias conclusions because grid decarbonisation can materially alter the estimated benefit of an intervention over time. Prospective LCA addresses this issue through structured scenario building and prospective inventory modelling, including approaches that couple integrated assessment model (IAM) scenarios to background databases, generate prospective LCI databases, and provide

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broader methodological guidance (Mendoza Beltran et al., 2020; Sacchi et al., 2022; Thonemann et al., 2020). Second, prospective inventory models contain explicit future assumptions and therefore benefit from structured scenario-based inventory modelling (Langkau et al., 2023). Third, applied LCAs do not always propagate quantitative uncertainty explicitly (Lloyd & Ries, 2007), and interpretation and priority setting become weaker when the main drivers of uncertainty remain unclear (Patouillard et al., 2019).

Recent building decarbonisation research spans mitigation pathways, future electricity scenarios, and dynamic or whole-life carbon assessment. Pathway studies examine demand-side strategies and decarbonisation trajectories across countries or sectors (Camarasa et al., 2022; van Heerden et al., 2025). Although not directly comparable, building-level assessments have also examined low-energy or passive-house cases under current and future electricity-mix scenarios (Norouzi et al., 2022). Other work extends this perspective to dynamic or whole-life carbon under changing backgrounds. Kordilas et al. (2025) assessed evolving grid and material mixes in a London office case. Norouzi et al. (2025) modelled temporal grid and material emission factors for passive houses in the United Kingdom. McLaren et al. (2025) showed why time-differentiated climate analysis also matters for building elements in New Zealand. Taken together, these studies show that future backgrounds can materially alter building-level results. However, they mainly inform design choices, pathway analysis, or long-horizon assessment, whereas campus managers need repeatable annual comparisons of operator-controlled levers. We therefore analyse energy and waste together as two routine operational levers at building-year resolution. The focus is on annual operational comparisons for a fixed, audited building rather than on whole-life design optimisation or dynamic simulation.

This study operationalises established prospective-LCA elements for campus operations in a form designed for routine building-level decision support. It contributes a practical workflow that improves scenario traceability, clarifies small operational differences, and supports reuse across comparable buildings and reporting years.

Against this background, this study addresses three research questions (RQs). RQ1 asks how much a future Queensland electricity background reduces annual operational 100-year global warming potential (GWP100; hereafter GWP) under a fixed foreground. RQ2 asks how large the waste-diversion effect is relative to grid decarbonisation. RQ3 asks which parameters dominate uncertainty in these contrasts.

2 Methods

This study adhered to International Organization for Standardization (ISO) 14040/14044 (ISO, 2006a; b). We used the Australian Life Cycle Inventory (AusLCI) v1.42 release in openLCA 2.5 (Ciroth, 2007). openLCA was the primary platform for life cycle inventory (LCI) and life cycle impact assessment (LCIA) calculations. Climate change impacts were characterised with the Intergovernmental Panel on Climate Change (IPCC) 2013 GWP100 method (IPCC, 2014). Brightway2 2.47 (Mutel, 2017) in Python 3.11.x was used as a diagnostic metamodel for paired contrasts and screening sensitivity analysis. Abbreviations, parameter definitions, unit factors, Monte Carlo settings, and supporting screenshots are provided in Supplementary Information S11. Units and rounding conventions were harmonised across scenarios.

2.1 Goal, scope, functional unit, reference flow and boundary

The study aimed to quantify annual operational climate impacts for a university building and to support operational decision-making. Energy and waste were analysed together because they are the two routine operational levers available at building-year resolution for this asset. The analysis combined openLCA-based reporting with a Brightway2 diagnostic metamodel. An attributional perspective was adopted (Ekvall & Weidema, 2004), with market-mediated consequential effects excluded from the baseline and examined only in targeted sensitivity runs where relevant. Following the terminology recommendations in Arvidsson et al. (2024), the audited 2024 foreground was held fixed while future-consistent background electricity scenarios were varied. The 2×2 scenario design varied two levers: electricity background and diversion rate (d), consistent with recent guidance on scenario-based inventory modelling (Langkau et al., 2023).

The case study was the Advanced Engineering Building (AEB) at The University of Queensland, St Lucia, Queensland, Australia, with 2024 as the operational year. The 2×2 design compared two electricity backgrounds (the current Queensland background and Queensland-2050) and two diversion rates (approximately 64% and 77%); accordingly, the case setting, electricity background assumptions, and waste-management modelling are specific to Queensland. The overall two-layer workflow and the role of each modelling environment are shown in Fig. 1a.

The functional unit (FU) was one building-year of AEB operations, aligning the inventory with metered annual

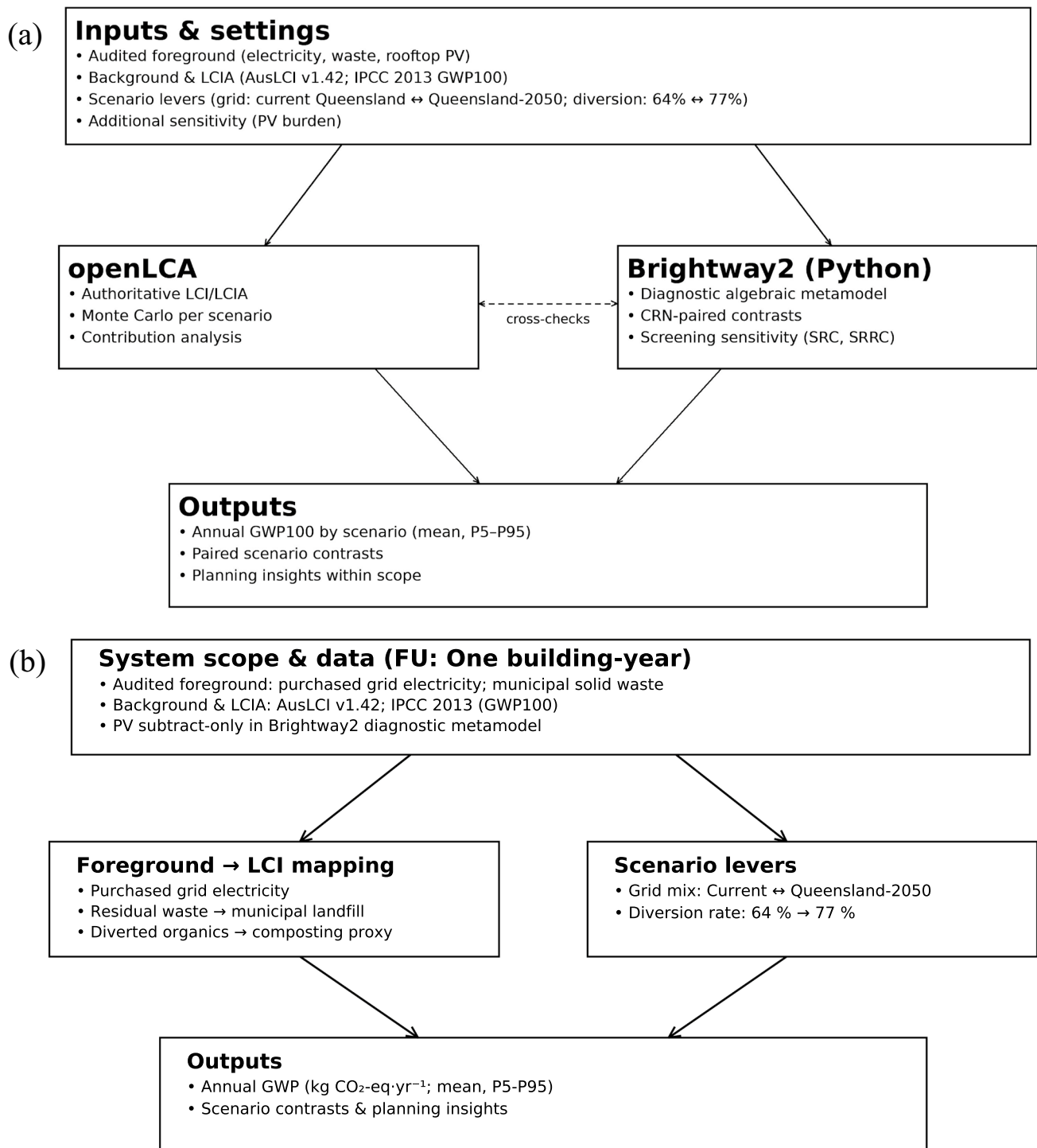


Fig. 1 **a** Two-layer workflow used in this study. The upper box summarises the audited foreground and the background settings. It also shows the life cycle impact assessment (LCIA) settings, the main scenario levers, and the photovoltaic (PV) burden sensitivity. The main levers are the electricity background and the diversion rate (d). The left box shows openLCA. It provides life cycle inventory (LCI) and LCIA results, per-scenario Monte Carlo runs, and contribution analysis. The right box shows Brightway2, which is used only as a diagnostic algebraic metamodel to support paired contrasts using common random numbers, sensitivity screening, and cross-checks against

openLCA. The dashed two-way arrow denotes these cross-checks. The lower box summarises the reported outputs. These outputs include annual 100-year global warming potential (GWP), percentile ranges, and paired contrasts. **b** System scope, foreground mapping, and scenario levers for the one-building-year functional unit. The audited foreground includes purchased grid electricity and municipal solid waste, with residual waste mapped to landfill and diverted organics to a composting proxy. In the Brightway2 diagnostic metamodel, PV is represented as subtract-only. Main scenario levers are the electricity background and the diversion rate (d)

activity and supporting within-asset operational decision-making. Area-normalised intensities, expressed as $\text{kg CO}_2\text{-eq m}^{-2} \text{ yr}^{-1}$, are also reported for contextual benchmarking across broadly comparable institutional buildings. The foreground system included three purchased processes: grid electricity, municipal landfill, and a composting proxy for diverted organics. The reference flow for electricity was annual building electricity demand, E_{tot} . It was calculated from energy use intensity (EUI) and floor area (A).

$$E_{\text{tot}} = \text{EUI} \times A \quad (1)$$

In Eq. (1), EUI was set to about $120 \text{ kWh m}^{-2} \text{ yr}^{-1}$ and A to $21,000 \text{ m}^2$, yielding E_{tot} of about $2.52 \times 10^6 \text{ kWh yr}^{-1}$ (Supplementary Information S11, Table S1). The waste reference flow was annual municipal solid-waste mass (W_{tot}).

The reported analysis covered audited annual building operations and was implemented as a gate-to-gate operational assessment within the use stage. The reported baseline

included purchased grid electricity and municipal solid-waste management, with the latter represented by landfill and a composting proxy. Construction, maintenance and renewal, water and wastewater services, occupant transport, and end-of-life services were excluded. An annualised materials link was retained only for boundary auditing and was deactivated in the reported scenarios; accordingly, the annualised materials term (BASE) was set to 0 in Eq. (5). Annual municipal solid-waste mass was partitioned by the diversion rate (d), with diverted organics routed to compost. In the reported baseline, diverted non-organics received no avoided-burden credit ($\text{CF}_{\text{divert_alt}} = 0$), whereas alternative recycling or energy-recovery credits were examined only in sensitivity runs and kept separate from the baseline results. Within the declared gate-to-gate boundary, a 5% completeness threshold, consistent with ISO 14044 (ISO, 2006b), was verified by contribution analysis for all scenarios; omitted foreground and background contributions did not exceed 5% of functional-unit GWP. Table 1 summarises

Table 1 System boundary and inclusion/exclusion summary for the gate-to-gate operational assessment

Process or flow	Included in reported baseline?	Baseline representation	Reason	Notes/sensitivity treatment
Purchased grid electricity	Included	Queensland current or Queensland-2050 electricity background	Dominant operational energy input and primary comparison lever	Compared across prospective electricity backgrounds
Rooftop photovoltaic (PV) generation	No (diagnostic only)	Diagnostic only ($\text{CF}_{\text{PV}} = 0$)	Preserves auditable reported totals	Subtract-only in the Brightway2 diagnostic metamodel
Municipal solid waste to landfill	Included	AusLCI/openLCA municipal landfill dataset	Core operational waste-management pathway	Annual waste mass partitioned by diversion rate d
Diverted organics	Included	Composting proxy	Represents an auditable diverted organic waste stream	Alternative CH_4 GWP100 setting tested in sensitivity runs
Diverted non-organics	Included without avoided-burden credit	$\text{CF}_{\text{divert_alt}} = 0$ in baseline	Conservative baseline; avoids embedding recycling or energy-recovery credits in core results	Recycling or energy-recovery credits examined only in sensitivity runs
Construction of building	Excluded	Not modelled in reported scenarios	Study focuses on the operation phase only	Outside gate-to-gate scope
Maintenance and renewal	Excluded	Not modelled in reported scenarios	Outside operational scope	Could be added in a future whole-life assessment
Water and wastewater services	Excluded	Not modelled in reported scenarios	Outside current study focus and unavailable in the audited foreground	Outside current boundary
Occupant transport	Excluded	Not modelled in reported scenarios	Outside the building operational foreground	A separate mobility analysis would be required
End-of-life services	Excluded	Not modelled in reported scenarios	Outside operational scope	Outside gate-to-gate boundary
Annualised materials link	Retained for boundary audit only	$\text{BASE} = 0$ in reported scenarios	Boundary check only; not part of reported baseline totals	Preserved for transparent expansion in future work

The table summarises the reported baseline boundary. $\text{CF}_{\text{divert_alt}}$ denotes the avoided-burden credit for diverted non-organics. BASE denotes the annualised materials term. Exploratory credits and alternative parameter settings are reported separately as sensitivities. They are not mixed with the baseline results

the system boundary, baseline treatment, and sensitivity handling. Figure 1b summarises the system scope, foreground mapping, and scenario levers for the functional unit of one building-year.

Operationally, the waste scenario represented a higher service-level diversion case rather than a new treatment technology. The same audited annual waste mass was reallocated from $d=0.64$ to $d=0.77$. At the central values reported in Supplementary Information S11 (Table S1), W_{tot} was about 50,000 kg yr⁻¹ and the organic share of diverted waste (s_{org}) was about 0.60, so the reallocation corresponded to about 6.5 t yr⁻¹ of additional diverted material. Of this shift, about 3.9 t yr⁻¹ moved from landfill to compost and about 2.6 t yr⁻¹ became additional non-organic diversion. Because the building-level audit resolved d and the organic share more reliably than material-specific downstream recycling pathways, the waste component was modelled as a service-level diversion scenario.

2.2 LCI data, LCIA method and data quality

For grid electricity, we used two named AusLCI datasets in openLCA. The current background was “electricity, low voltage, Queensland/AU U.” The Queensland-2050 background was “electricity mix, 2050 Queensland/AU U”. The study did not generate a new prospective database or perform IAM-specific pathway mapping.

For waste management, we used the datasets “waste treatment, municipal waste average, at landfill” and “composting of organics (proxy).” This representation matches the available waste-audit categories and preserves a transparent operational foreground. The graphical user interface (GUI) unit factor for landfill was 0.833 kg CO₂-eq kg⁻¹. For readability, CF denotes the relevant emission factor in the notation that follows. In Eq. (5), CF_{landfill} denotes the landfill emission factor, whereas CF_{divert_eff} denotes the effective emission factor for the diverted fraction defined in Eq. (2). CF_{compost} denotes the composting factor for diverted organics and was set to 0.046 kg CO₂-eq kg⁻¹ (Supplementary Information S11, Table S2). Documentation followed the AusLCI Manual v1.40 (Grant & Rocha, 2022).

For uncertainty analysis, metered electricity and audited municipal solid-waste flows were used to define input distributions. Sampled inputs included energy use intensity (EUI), annual municipal solid-waste mass (W_{tot}), installed photovoltaic capacity (PV_{kWp}), photovoltaic yield (PV_{yield}), the photovoltaic self-use fraction (p_{PV}), and the organic share of diverted waste (s_{org}). By contrast, the diversion rate (d) was fixed by scenario at 0.64 or 0.77 in the openLCA Monte Carlo runs. In the Brightway2 diagnostic metamodel, annual self-used on-site photovoltaic generation (PV_{gen}) was treated as

a derived variable (Sect. 2.3). The full parameter table is given in Supplementary Information S11 (Table S1).

$$CF_{\text{divert_eff}} = \text{sorg} \times CF_{\text{compost}} + (1 - \text{sorg}) \times CF_{\text{divert_alt}} \quad (2)$$

In Eq. (2), the avoided-burden credit for diverted non-organics is denoted CF_{divert_alt} and equals 0 in the baseline. Equation (2) also defines the effective factor applied to the diverted fraction, denoted CF_{divert_eff} in Eq. (5). Recycling or energy-recovery credits for diverted non-organics were examined only in sensitivity runs and were not mixed with baseline reporting.

Cross-tool agreement is reported in Sect. 3.3 and Supplementary Information S11. Key parameters, especially EUI and W_{tot} , were screened with a simplified pedigree approach covering temporal, geographical, and technological representativeness as well as methodological rigour. Metered electricity data were rated as high quality, whereas audit-based waste data were rated as moderate quality. These ratings informed the uncertainty ranges in Supplementary Information S11 (Table S1). Waste distributions were assigned larger geometric standard deviations than metered electricity. This choice followed empirically based pedigree factors (Ciroth et al., 2016).

For climate change, characterisation factors were treated as fixed, and parameter uncertainty was propagated on the inventory side. This is consistent with probabilistic LCA practice (Huijbregts et al., 2001; Lloyd & Ries, 2007). Electricity impacts were reported on a location-based basis using the AusLCI GUI factors. Market-based adjustments, including residual mixes, energy attribute certificates (EACs), and power purchase agreements (PPAs), were not applied. This choice preserved alignment with the physically delivered regional mix and avoided double counting at the campus scale.

The installed rooftop photovoltaic (PV) system capacity was 95.75 kWp (The University of Queensland, n.d.). In the Brightway2 diagnostic metamodel, rooftop PV was treated as subtract-only (CF_{PV}=0). A separate sensitivity run applied CF_{PV}=0.045 kg CO₂-eq·kWh⁻¹, with p_{PV} varied uniformly from 0.85 to 1.00 (Supplementary Information S11, Table S1). For composting, the baseline factor was CF_{compost}=0.046 kg CO₂-eq·kg⁻¹, corresponding to CH₄ GWP100=28 without climate-carbon feedback (IPCC, 2014); a sensitivity run used CH₄ GWP100=34, increasing CF_{compost} to 0.056 kg CO₂-eq·kg⁻¹ (Supplementary Information S11, Table S2). Table 2 summarises the scenario inputs, baseline settings, and sensitivity treatments.

2.3 Scenario design, uncertainty, sensitivity and algebraic metamodel

The workflow was implemented across two complementary environments. openLCA stored the named datasets, performed scenario-specific Monte Carlo runs, and generated

Table 2 Scenario input summary for the prospective operational assessment

Input category	Variable or setting	Reported baseline or current setting	Alternative, scenario level, or sensitivity treatment
Scenario lever	CF_grid	Current Queensland background; 0.851 kg CO ₂ -eq·kWh ⁻¹	Queensland-2050 background; 0.445 kg CO ₂ -eq·kWh ⁻¹
Scenario lever	d	about 64% diversion	77% diversion
Audited foreground	EUI	Sampled around the metered 2024 building value (about 120 kWh m ⁻² yr ⁻¹)	Held common across scenarios
Audited foreground	A	21,000 m ²	Fixed floor area
Audited foreground	W_tot	Sampled from audited 2024 municipal solid-waste (MSW) totals	Held common across scenarios
Audited foreground	s_org	Sampled organic share of diverted waste	Held common across scenarios
On-site generation	PV_kWp	Lognormal distribution around the installed rooftop PV capacity (GM=95.75 kWp; GSD=1.025)	Sampled rooftop PV capacity; annual self-used PV generation enters as derived PV_gen
On-site generation	PV_yield	Triangular distribution (1200 / 1350 / 1620 kWh kWp ⁻¹ yr ⁻¹)	Sampled PV yield; annual self-used PV generation enters as derived PV_gen
On-site generation	p_PV	Uniform distribution from 0.85 to 1.00	Sampled self-use fraction; enters derived PV_gen
Baseline accounting	CF_divert_alt	0 in baseline	Recycling or energy-recovery credit in sensitivity only
Baseline accounting	CF_PV	0 (Brightway2 diagnostic only)	0.045 kg CO ₂ -eq kWh ⁻¹ in sensitivity
Baseline accounting	CF_compost	0.046 kg CO ₂ -eq·kg ⁻¹ (CH ₄ GWP100=28)	0.056 kg CO ₂ -eq kg ⁻¹ when CH ₄ GWP100=34
Uncertainty setting	openLCA Monte Carlo	n = 2,000 per scenario	Independent sampling
Uncertainty setting	Brightway2 paired runs	n = 3,000	Paired contrasts using common random numbers

d diversion rate, *EUI* energy use intensity, *A* floor area, *W_tot* annual municipal solid-waste mass, *s_org* organic share of diverted waste, *PV_kWp* installed photovoltaic capacity, *PV_yield* photovoltaic yield, *p_PV* photovoltaic self-use fraction, *PV_gen* annual self-used on-site photovoltaic generation, *CF_grid* grid electricity factor, *CF_divert_alt* avoided-burden credit for diverted non-organics, *CF_PV* photovoltaic life-cycle factor, *CF_compost* composting factor, *GM* geometric mean, *GSD* geometric standard deviation

Full parameter definitions and distributions are provided in Supplementary Information S11 (Tables S1–S3)

the reported LCI and LCIA results, whereas Brightway2 represented the same audited foreground algebraically to support paired contrasts and screening sensitivity analysis. Brightway2 thus complemented openLCA without replacing it as the basis for reported results.

In practice, openLCA provided per-scenario Monte Carlo results under independent sampling, whereas Brightway2 implemented CRN-based paired contrasts and screening sensitivity analyses with the audited foreground held fixed (Supplementary Information S11, Table S3). Monte Carlo simulation used $n = 2,000$ iterations per scenario in openLCA and $n = 3,000$ iterations in Brightway2 for paired contrasts. Ignoring dependence can bias variance estimates and mis-rank sensitivities (Groen & Heijungs, 2017).

Global sensitivity analysis used standardised regression coefficients (SRCs) and standardised rank regression coefficients (SRRCs) as a first-order screen because model behaviour was near-linear. The sign of β indicates effect direction, and its absolute value indicates relative importance.

The reduced Brightway2 screen varied EUI, *W_tot*, *p_PV*, and *d*, with panels centred on the lower-diversion scenario.

The Brightway2 metamodel represents annual GWP algebraically using the notation defined in Supplementary Information S11 (Table S-Glossary and Table S1); the key sampled inputs are summarised in Table 2. Equations (3–5) are included to make the model structure explicit and should be read as diagnostic expressions rather than replacements for the openLCA calculations. In Eq. (3), *PV_gen* denotes annual self-used on-site photovoltaic generation, derived from *PV_kWp*, *PV_yield*, and *p_PV*. In Eq. (4), *f_PV* denotes the fraction of annual electricity demand met by self-used PV. In Eq. (5), *CF_landfill* denotes the landfill emission factor, and *CF_divert_eff* denotes the effective emission factor for the diverted fraction defined in Eq. (2).

$$PV_gen = PV_kWp \times PV_yield \times p_PV \quad (3)$$

$$f_PV = \min(1, PV_gen/Etot) \quad (4)$$

$$\text{GWP} = \text{Etot} \times ((1 - f_{\text{PV}}) \times \text{CF}_{\text{grid}} + f_{\text{PV}} \times \text{CF}_{\text{PV}}) + \text{Wtot} \times [(1 - d) \times \text{CF}_{\text{landfill}} + d \times \text{CF}_{\text{divert_eff}}] + \text{BASE} \quad (5)$$

2.4 Reproducibility and quality assurance/quality control (QA/QC)

The openLCA project was version controlled, dataset versions (AusLCI v1.42) were archived, and high-precision scenario exports were stored for each scenario, including the mean, standard deviation (SD), and 5th to 95th percentiles (P5–P95). A minimal Python/Brightway2 package and supporting screenshots of foreground networks and dataset panels were also archived in Supplementary Information S11. Tables S0–S4 and Figs. S1–S3 document the scenario design, parameter settings, Monte Carlo configuration, and diagnostic checks.

Annual electricity demand was based on main-meter readings, rooftop PV output was derived from inverter logs, and waste quantities and diversion shares were aligned with the most recent audits and reconciled against monthly hauler records. Short gaps of 48 h or less were bridged with nearest-neighbour days matched to occupancy and degree-days, and outliers greater than 3 standard deviations were checked against maintenance logs. GUI-displayed AusLCI factors remained the basis for reporting.

The scenario design and analysis settings were archived, and the Brightway2 seed was fixed at 2025, supporting transparent and repeatable diagnostic comparisons under the stated software and dataset versions. Baseline accounting remained separate from sensitivity analyses to avoid double counting.

3 Results

3.1 Annual GWP across scenarios

Mean annual operational GWP was 2.151×10^6 kg CO₂-eq·yr⁻¹ under the current grid and fell to 1.136×10^6 and 1.133×10^6 kg CO₂-eq·yr⁻¹ under Queensland-2050.

The corresponding floor-area intensities declined from 102.4 to 54.1 and 54.0 kg CO₂-eq·m⁻²·yr⁻¹, indicating an overall reduction of about 1.0×10^6 kg CO₂-eq·yr⁻¹. Percentile ranges are reported in Table 3 and illustrated in Fig. 2.

Relative to the current Queensland background, Queensland-2050 reduced annual GWP by 1,015,000 kg CO₂-eq·yr⁻¹ at 64% diversion and by 1,018,000 kg CO₂-eq·yr⁻¹ at 77% diversion, corresponding to reductions of 47.2% and 47.3% (Table 4). By contrast, increasing diversion from about 64% to 77% changed annual GWP by only about 5.3×10^3 kg CO₂-eq·yr⁻¹ under both electricity backgrounds. This difference was smaller than the independent-sampling precision reported in Table 3 and was therefore more clearly resolved by CRN-based paired contrasts in Table 4.

3.2 Contributions and global sensitivity

Contribution analysis showed that purchased electricity dominated annual operational GWP, contributing about 2.1×10^6 kg CO₂-eq·yr⁻¹ under the current Queensland electricity background. By contrast, waste-related terms were on the order of 10^4 kg CO₂-eq·yr⁻¹ and thus roughly two orders of magnitude smaller. In the reduced Brightway2 sensitivity screen (Fig. 3), energy use intensity dominated variance under both electricity backgrounds ($\beta \approx +0.95$; $R^2 \approx 1.00$), the photovoltaic self-use fraction, p_{PV} , was secondary ($\beta \approx -0.06$), and W_{tot} and d remained minor. SRRCs were directionally consistent with SRCs and yielded the same sensitivity ranking; for brevity, Fig. 3 reports SRCs only. Accordingly, increasing diversion to 77% changed the waste term by only a few tonnes CO₂-eq·yr⁻¹ at current waste volumes.

Table 3 Results from openLCA Monte Carlo (n=2,000 per scenario; independent sampling)

Scenario	Mean (kg CO ₂ -eq·yr ⁻¹)	P5	Median	P95	SD	n
Current grid, 64% diversion	2,151,000	1,838,000	2,139,000	2,500,000	201,600	2,000
Current grid, 77% diversion	2,151,000	1,844,000	2,141,000	2,483,000	197,800	2,000
Queensland-2050 grid, 64% diversion	1,136,000	964,500	1,130,000	1,317,000	108,000	2,000
Queensland-2050 grid, 77% diversion	1,133,000	964,400	1,128,000	1,321,000	108,000	2,000

Units: kg CO₂-eq per building-year

P5 and P95 denote the 5th and 95th percentiles. SD denotes standard deviation. Table 3 reports openLCA Monte Carlo means under independent sampling, with n=2,000 per scenario. Values are rounded to the nearest 10³ kg. Pairwise changes smaller than about $(3-6) \times 10^3$ kg can be obscured at this precision

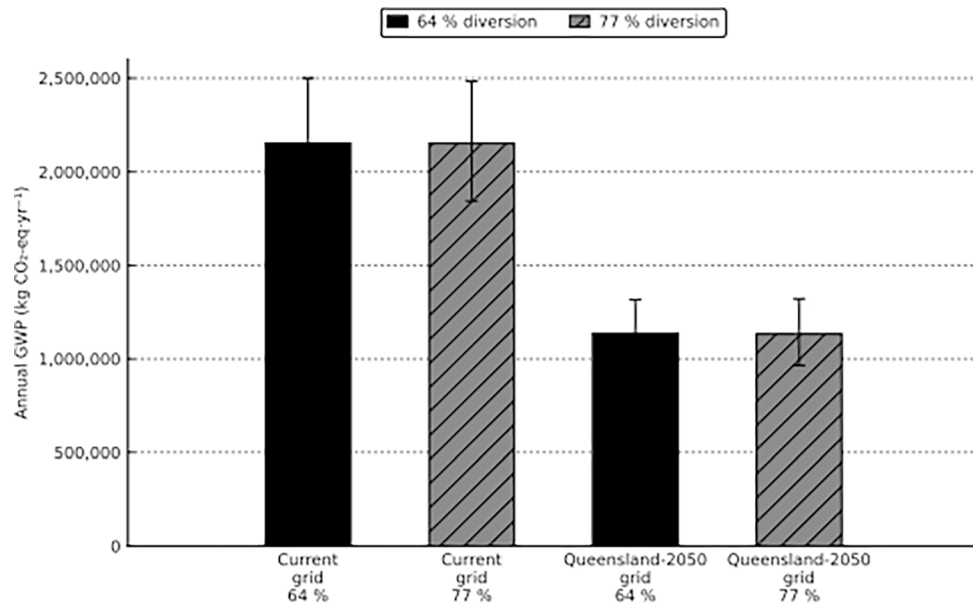


Fig. 2 Annual 100-year global warming potential (GWP) for the four scenarios. The y-axis reports annual GWP in kg CO₂-eq yr⁻¹. Solid black bars denote 64% diversion. Hatched grey bars denote 77% diversion. Each bar represents the openLCA Monte Carlo mean, with n=2,000. Whiskers indicate the interval from the 5th percentile (P5)

to the 95th percentile (P95). Scenario labels combine electricity background and diversion level. They are Current grid, 64% diversion; Current grid, 77% diversion; Queensland-2050 grid, 64% diversion; and Queensland-2050 grid, 77% diversion. Scenario definitions are provided in Supplementary Information SI1, Table S0

Table 4 Pairwise scenario contrasts in annual GWP

Pairwise contrast	Δ (kg CO ₂ -eq·yr ⁻¹)	Δ (%)	Method notes
Current grid, 77% diversion vs. Current grid, 64% diversion	$\approx - (5.3 \pm 0.1) \times 10^3$	$\approx - 0.24\%$	Equation (5) with GUI unit factors; paired precision improved with common random numbers
Queensland-2050 grid, 77% diversion vs. Queensland-2050 grid, 64% diversion	$\approx - (5.3 \pm 0.1) \times 10^3$	$\approx - 0.47\%$	Equation (5) with GUI unit factors; paired precision improved with common random numbers
Queensland-2050 grid, 64% diversion vs. Current grid, 64% diversion	- 1,015,000	- 47.2%	Equation (5) with GUI unit factors; paired precision improved with common random numbers
Queensland-2050 grid, 77% diversion vs. Current grid, 77% diversion	- 1,018,000	- 47.3%	Equation (5) with GUI unit factors; paired precision improved with common random numbers

Contrasts were calculated with Eq. (5) and GUI unit factors. Common random numbers were used, with seed=2025 and n=3,000. For the reported baseline contrasts in Tables 3–4, the no-PV reporting boundary was retained (CF_PV=0; BASE=0); Eqs. (3–5) are diagnostic expressions only, and PV-related terms were activated only in the Brightway2 sensitivity analyses, including Fig. 3. The waste-diversion contrasts are CRN-based paired diagnostic estimates and cannot be directly recovered by subtracting the independently sampled, rounded means reported in Table 3

3.3 Robustness across tools

Overall, cross-tool concordance was high, although minor residual differences between equation-level checks and openLCA scenario means reflected dataset-level boundary definitions. With the check-only alignment constant applied in the algebraic metamodel, scenario means agreed with openLCA within $\pm 0.5\%$. This adjustment was used solely for diagnostic comparison and did not redefine the reported results, which remained based on the openLCA GUI unit factors.

4 Discussion

4.1 Interpretation of the findings

Within the present boundary, annual operational GWP was dominated by electricity-related variables, so the largest reductions arose from lower energy use intensity and a lower-carbon electricity background rather than from higher diversion at current waste volumes. The scenario ranking was clear: grid decarbonisation had a much larger effect on annual operational GWP than the tested increase

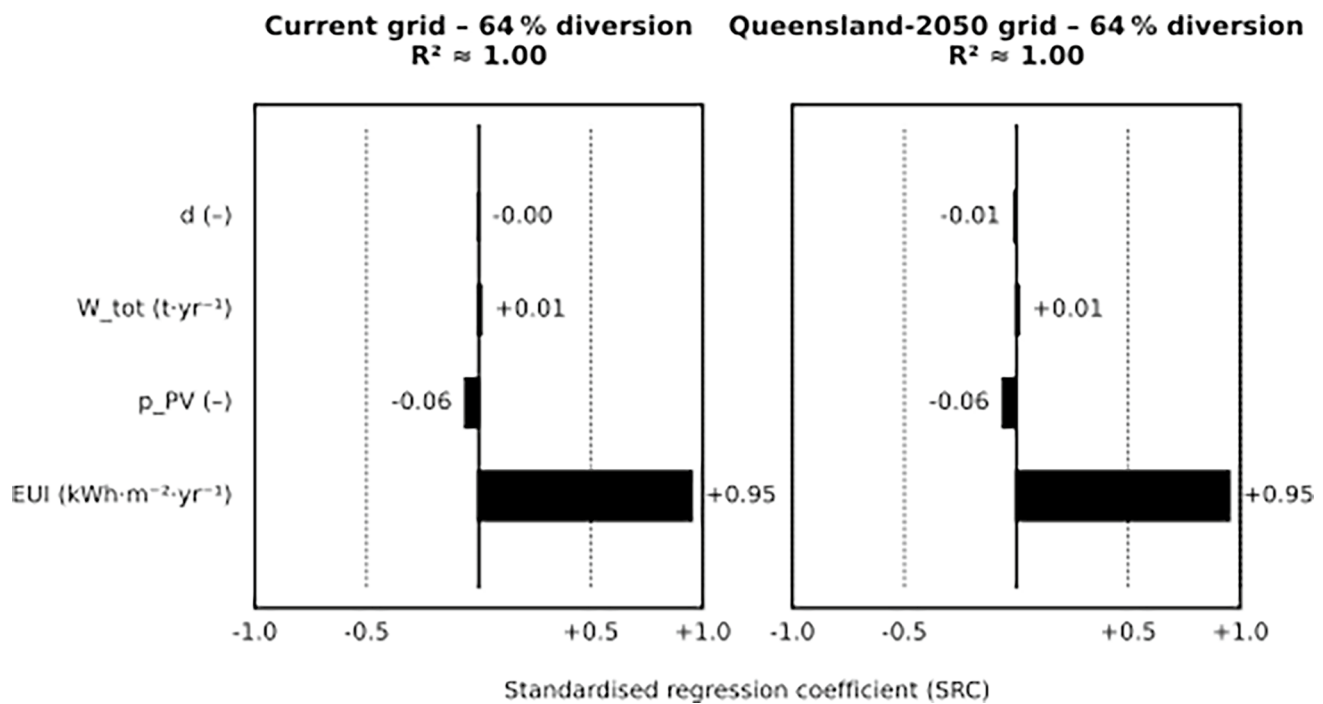


Fig. 3 Screening sensitivity expressed as standardised regression coefficients (SRCs). The figure shows the reduced Brightway2 diagnostic screening set. The left panel corresponds to the current grid, 64% diversion case, and the right panel to the Queensland-2050 grid, 64% diversion case. Inputs shown are the diversion rate (d), total waste mass (W_{tot}), photovoltaic self-use fraction (p_{PV}), and

energy use intensity (EUI). Common random numbers were used (seed=2025; $n=3,000$). Energy use intensity was dominant in both panels ($\beta \approx +0.95$). The photovoltaic self-use fraction was secondary ($\beta \approx -0.06$), whereas W_{tot} and d were minor, with coefficients close to zero. The coefficient of determination (R^2) was approximately 1.00 in both panels

in diversion (Tables 3 and 4). This ordering is consistent across both the contribution analysis and the sensitivity screening. Purchased electricity dominates the annual total, while waste-related terms remain much smaller in magnitude. Energy use intensity is the main driver of variance, the photovoltaic self-use fraction is secondary, and both W_{tot} and d remain minor. In practical terms, measures targeting electricity demand and electricity supply therefore offer larger near-term climate benefits than marginal increases in diversion.

The small waste-diversion effect does not imply that waste management is unimportant. Rather, for this building and boundary, annual climate leverage from waste remained limited relative to purchased electricity because audited waste volumes were small. This interpretation is consistent with campus LCA and carbon-footprint evidence showing that electricity often dominates emissions (Qiu & Dong, 2024; Wang et al., 2023), although university waste studies continue to identify substantial recycling potential in campus streams (Zhang et al., 2020). Waste therefore remains relevant for operational management and circularity, but its annual greenhouse-gas leverage was secondary in the present case.

4.2 Comparison with related literature and scenario governance

Within the higher education institution (HEI) and campus LCA literature reviewed here, comparable case studies at the university-building level appear to remain limited, and standardised assessment approaches are still emerging (Jürgens et al., 2023; Qiu & Dong, 2024). Recent reviews also identify temporal background modelling and uncertainty reporting as continuing weaknesses (Fnais et al., 2022; Qiu & Dong, 2024). Under the present gate-to-gate boundary, the results remained dominated by purchased electricity. Mean intensity declined from 102.4 to 54.1 and 54.0 kg CO₂-eq·m⁻²·yr⁻¹ when the electricity background shifted to Queensland-2050, corresponding to reductions of about 47%. This reduction is smaller than the 58–70% GWP decline reported for passive-house dwellings in Northern Ireland, but it remains directionally consistent with that study (Norouzi et al., 2022). Later UK work likewise shows that temporal grid and material decarbonisation can substantially reduce whole-life carbon (Norouzi et al., 2025). The difference in magnitude is unsurprising because the present case concerns a university building with a fixed audited foreground and a gate-to-gate operational scope, whereas

those studies examine low-energy dwellings and, in part, whole-life trajectories under different background assumptions. More broadly, the results support the conclusion that future electricity backgrounds can materially alter building results, although the precise magnitude remains case- and scope-dependent.

Prospective LCA research has increasingly shifted from static baselines toward future-consistent background modelling. Early methodological work showed how IAM scenarios can be coupled to background databases and operationalised in prospective life cycle inventory databases (Mendoza Beltran et al., 2020; Sacchi et al., 2022). Later guidance clarified broader good practice and the conditions for wider application of prospective life cycle inventory databases (Steubing et al., 2023; Thonemann et al., 2020). Against this background, the present study contributes an operations-centred workflow. The foreground is fixed, background datasets are named and versioned, and scenario contrasts can be reproduced without redefining the core LCI and LCIA model. This workflow is consistent with scenario-based inventory modelling guidance for prospective LCA, particularly its emphasis on transparent and consistent handling of future assumptions (Langkau et al., 2023).

The uncertainty treatment is consistent with current LCA practice, in which sensitivity analysis is used to prioritise modelling effort and to identify drivers of variance (Jaxa-Rozen et al., 2021; Patouillard et al., 2019). Given the near-linear behaviour observed here, SRCs and SRRCs were used as a first-order screen, while the algebraic metamodel was used to clarify model structure and paired contrasts. This role aligns with the transparency goals of symbolic-calculus approaches in LCA (Jolivet et al., 2021) and improves interpretability in prospective operational building LCA.

Within the broader digital twin literature, the present workflow sits at the lifecycle-information and scenario-analysis layer rather than the cyber-physical control layer. Manufacturing digital twins often emphasise operational data integration, learning, and prediction, whereas infrastructure and building-energy twins more directly connect virtual models to real-time data for coordination, monitoring, diagnosis, or control (Hwang et al., 2025; Mortlock et al., 2022; Moshood et al., 2024). By contrast, the present workflow keeps the physical asset outside a real-time control loop and instead links a fixed audited foreground to named background LCI scenarios, explicit uncertainty settings, and repeatable paired contrasts. In that sense, it may be viewed as a lifecycle-scenario representation for campus building operations, consistent with digital lifecycle twin discussions that emphasise lifecycle-oriented information management rather than real-time control (Riedelsheimer et al., 2021). Its value lies in governed reuse across buildings or reporting years while preserving dataset provenance and scenario definitions.

4.3 Limitations and uncertainty

First, the analysis reports GWP only; other environmental impact categories may respond differently as electricity backgrounds change, so multi-category follow-up assessment is warranted. These categories were excluded because the present decision problem concerned annual greenhouse-gas prioritisation rather than a full multi-category comparison. Given the dominance of purchased electricity within the present gate-to-gate scope, the main scenario ranking is unlikely to reverse under the stated boundary conditions. Nevertheless, as electricity-related climate burdens decline, broader impact trade-offs may become more decision-relevant in building LCA (Fnais et al., 2022; Kordilas et al., 2025).

Second, uncertainty was propagated on the inventory side while characterisation factors were treated as fixed, a common choice in probabilistic LCA that nevertheless omits model uncertainty in LCIA (Huijbregts et al., 2001; Lloyd & Ries, 2007). Within each scenario, sampled inputs were treated as independent, except that matched contrasts were paired using common random numbers. This assumption simplifies the treatment of potential real-world correlations among energy demand, PV performance, and waste generation and may therefore affect estimated uncertainty widths. However, it is unlikely to reverse the ranking because EUI overwhelmingly dominated first-order sensitivity.

Third, the reported boundary is gate-to-gate and operation-centred. Construction, refurbishment, water and wastewater services, occupant transport, and end-of-life services were excluded. These stages should be assessed separately when embodied impacts or off-site services drive decisions. The accounting choices were intentionally conservative: diverted non-organics received no avoided-burden credit in the reported baseline, whereas rooftop PV was treated as subtract-only in the Brightway2 diagnostic metamodel. These choices were intended to avoid double counting, but they may limit transferability to settings with explicit recycling pathways or higher PV penetration. In the present case, adding a PV life-cycle factor changed annual totals by less than 1%. Published PV LCA guidance supports transparent reporting of such choices and sensitivity testing (Frischknecht et al., 2020; Smith et al., 2024).

The zero-credit treatment for diverted non-organics should be understood as a conservative accounting baseline rather than as a claim that recycling never yields benefits. At the present waste scale, even more generous additional credits would remain small relative to the grid effect, so the ranking of levers would not be expected to change in this case.

The functional unit of one building-year is appropriate for within-asset operational decision support, but not for direct public comparison across buildings unless floor area,

occupancy intensity, service level, and use profile are comparable. The area-normalised results should therefore be read as secondary normalisations for contextual benchmarking rather than as stand-alone comparative claims. Transferability is likely to be highest for electricity-dominated institutional buildings with audited annual energy and waste data and comparable regional electricity and waste-management conditions. Buildings with larger waste volumes, different recycling credits, or higher PV penetration may show a different balance between levers. More broadly, HEI LCA studies continue to identify data availability as a major constraint on both holistic comparison and transferability (Jürgens et al., 2023; Schmidt et al., 2025).

4.4 Implications for operations, policy and method

Within the present boundary, the largest operational decarbonisation gains are associated with lower energy use intensity and a lower-carbon electricity background. Higher diversion remains important for circularity, stewardship, and compliance, but its annual climate effect is smaller than that of electricity-centred measures in this case. For operational reporting, a location-based electricity perspective can help maintain consistency across institutional buildings and sub-national reporting scales. Where market-based claims are also made, location-based and market-based results are best disclosed side by side and reconciled carefully. Procurement and audit guidance would likewise benefit from clear conditions for prospective LCI backgrounds, including provenance, versioning, and dataset specification. In practical terms, this points to demand-side efficiency measures such as recommissioning support and EUI-oriented performance targets.

The scenario structure also supports procurement and retrofit planning. It allows decision-makers to assess how electricity decarbonisation and efficiency improvements move institutional buildings toward organisation-wide targets and to represent operational risk through distributions rather than single values.

5 Conclusions

This study addressed three research questions on prospective operational decarbonisation in a university building. For RQ1, a future Queensland electricity background substantially reduced annual operational GWP relative to the current background, by about 47% under the fixed foreground examined here. For RQ2, increasing municipal waste diversion from about 64% to 77% produced only a small climate benefit at current audited waste volumes. Under the present conditions, electricity-centred measures therefore offer much larger near-term greenhouse-gas reductions than higher

diversion rates. For RQ3, the contribution and sensitivity analyses showed a stable ordering of influence across scenarios and sensitivities, including the PV life-cycle sensitivity. Purchased electricity dominated annual operational GWP, energy use intensity was the main driver of variance, p_{PV} was secondary, and W_{tot} and d remained minor in the reduced Brightway2 screen.

These findings indicate that, within the present attributional and gate-to-gate scope, institutions are likely to achieve larger near-term operational decarbonisation gains through lower energy use intensity and lower-carbon electricity backgrounds than through marginal increases in diversion alone. High diversion nevertheless remains important for circularity and operational stewardship, even though its annual climate effect is relatively small in this case study. More broadly, the study shows that prospective operational LCA can support routine building-management decisions when the foreground is fixed, background scenarios are explicitly defined, and uncertainty is reported transparently.

These conclusions apply to an attributional, gate-to-gate assessment of purchased electricity and municipal solid-waste management in an electricity-dominated institutional building and should therefore be transferred cautiously beyond similar operational and regional contexts.

6 Summary:

Supplementary Information S11

This supporting information provides model parameters, methodological details, uncertainty settings, and reproducibility materials for the life cycle assessment conducted in this study. Table S-Glossary defines abbreviations and symbols used throughout the manuscript and aligns notation for cross-tool concordance. Table S0 presents the 2×2 scenario design varying electricity background and waste diversion rate. Tables S1S3 compile parameter settings and Monte Carlo configurations for the foreground system and the Brightway2 diagnostic metamodel. Table S4 reports cross-tool concordance between openLCA and Brightway2 results. Figures S1S3 illustrate the system boundary, foreground mapping, and electricity background datasets.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s44498-026-00147-4>.

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Data availability High-precision scenario exports and the minimal Brightway2 metamodel are publicly available on Zenodo (<https://doi.org/10.5281/zenodo.19420738>). These materials are released under a Creative Commons Attribution 4.0 (CC BY 4.0) licence. They include the per-scenario Monte Carlo outputs and the scripts needed to reproduce the published values.

Code availability The minimal Brightway2 metamodel code is archived together with the data in the Zenodo repository cited above.

Declarations

Competing interests The authors declare no competing interests.

Ethics approval This article does not contain studies with human participants or animals performed by the authors. Ethics approval was therefore not required.

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